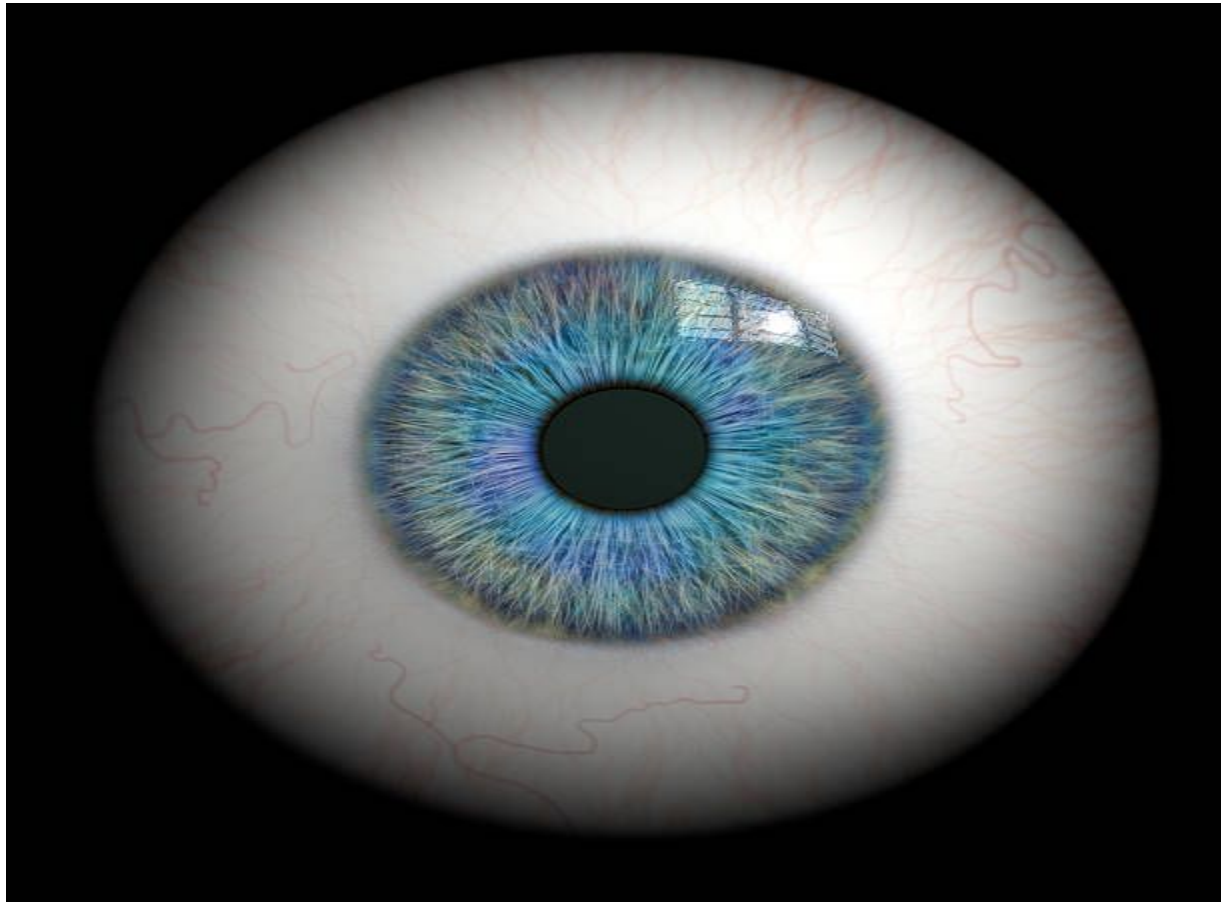
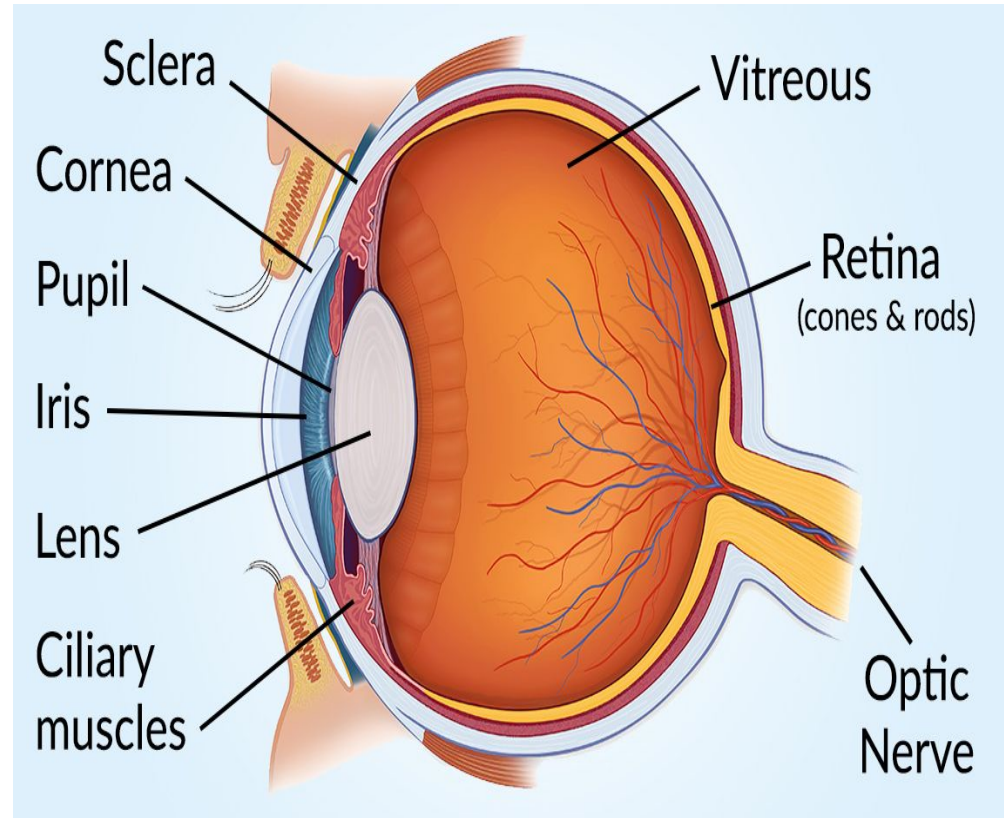


The Eyeball & Optic Nerve (II)



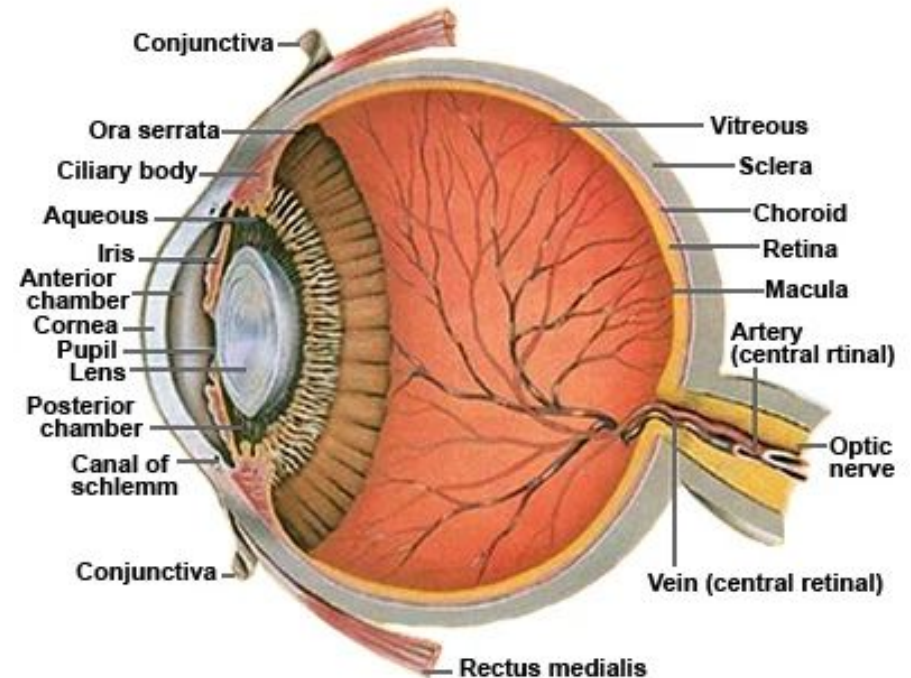
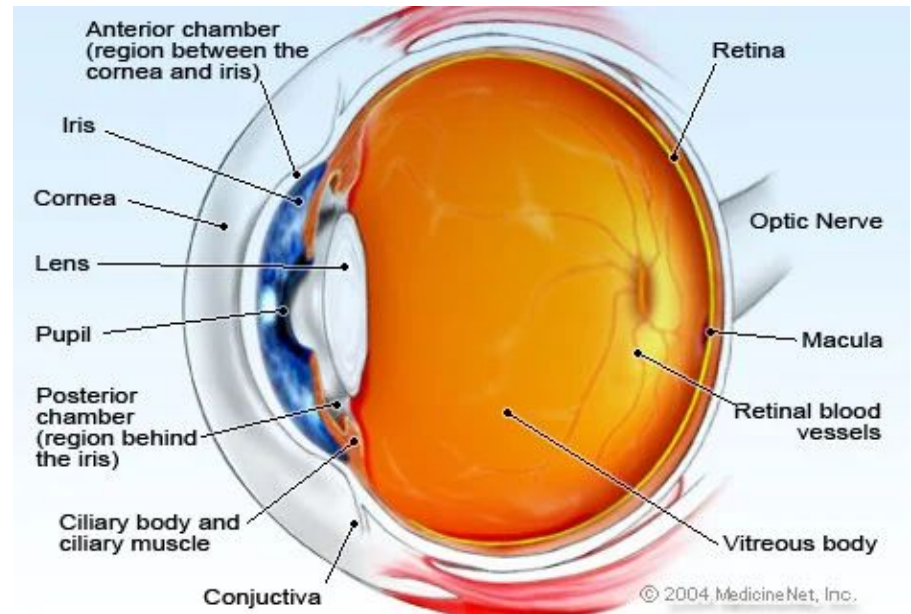
Introduction:

- The eyeball is the organ of sight.
- It is spherical in shape having a diameter of about 2.5 cm.
- It is made up of 3 concentric coats:
 1. Outer/fibrous coat: contains **sclera & cornea**
 2. Middle/vascular coat: also called uveal tract & comprises of **choroid, ciliary body & iris**.
 3. Inner/nervous coat is **retina**.



The outer coat:

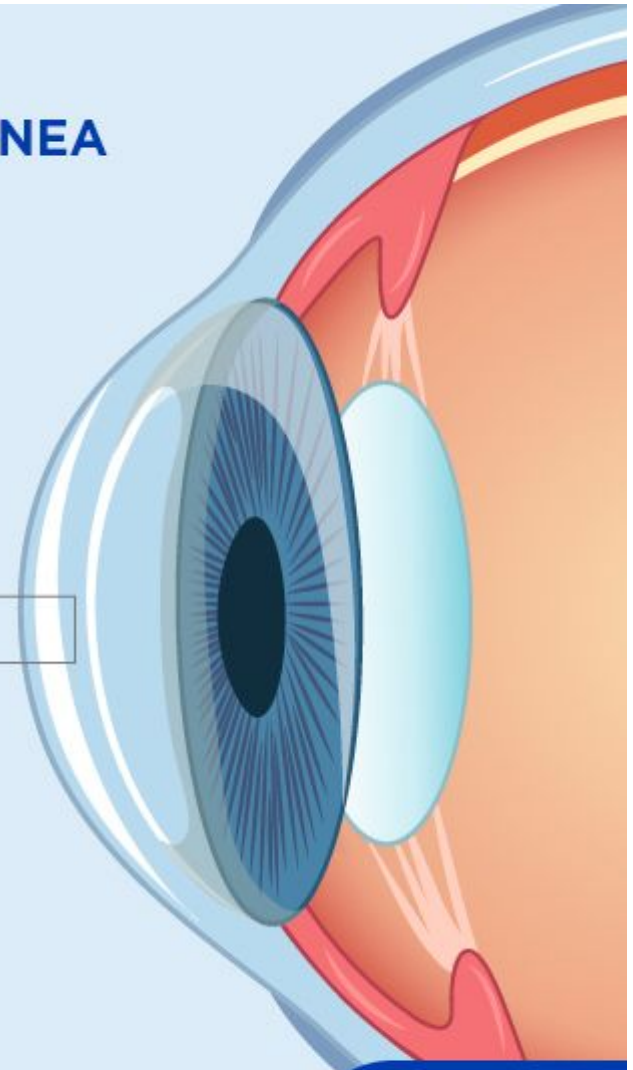
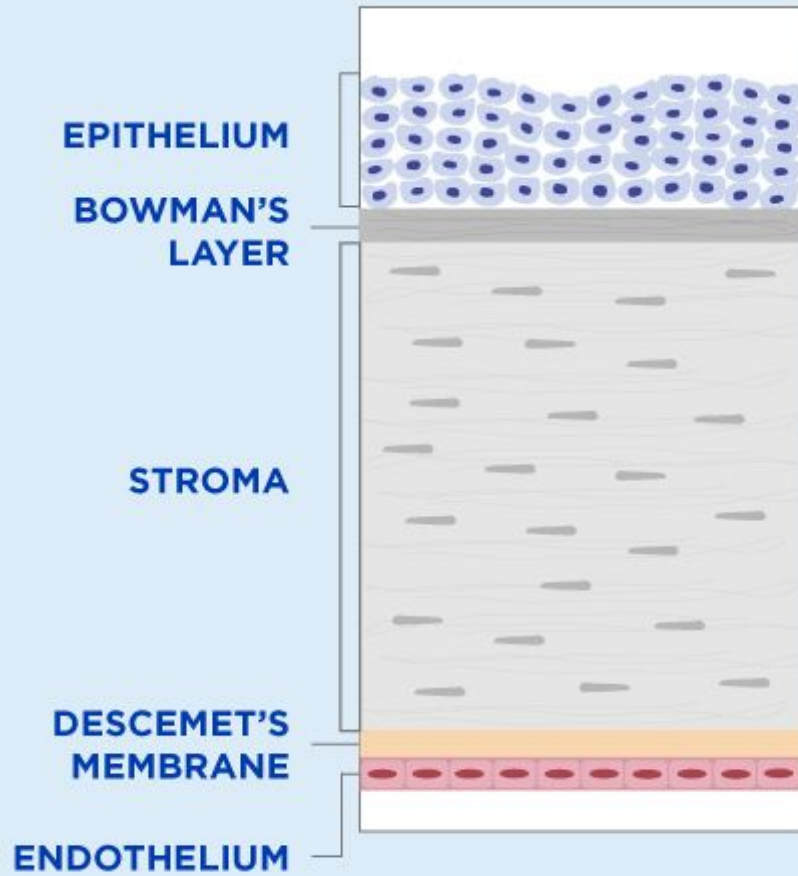
- **Sclera:** its outer surface is white, it is opaque & forms posterior five-sixth of the eyeball.
- Composed of dense fibrous tissue which maintains the shape of eyeball.
- It is weakest at the entrance of optic nerve where it shows numerous perforations called **lamina cribrosa** for the passage of optic nerve fibres.
- It is continuous anteriorly with the cornea at **sclerocorneal junction/limbus**.
- It is covered by Tenon's capsule which is a thin, loose membranous sheath around the eyeball extending from optic nerve to sclerocorneal junction.
- The deep part of limbus contains a circular canal called **canal of Schlemm**, through which aqueous humor is drained.
- Sclera is avascular and fused posteriorly with the dural sheath of optic nerve.



The outer coat:

- **Cornea:** is transparent and replaces the sclera over anterior one-sixth of the eyeball.
- It is more convex than sclera but the curvature diminishes with age.
- It is separated from iris by a space called anterior chamber of the eye.
- It is avascular but pain sensitive.
- Histologically cornea consists of the following layers from before backwards:
 1. Stratified squamous non-keratinized epithelium/corneal epithelium
 2. Bowman's membrane/anterior elastic lamina
 3. Substantia propria
 4. Descemet's membrane/posterior elastic lamina
 5. Simple squamous mesothelium

STRUCTURE OF THE CORNEA

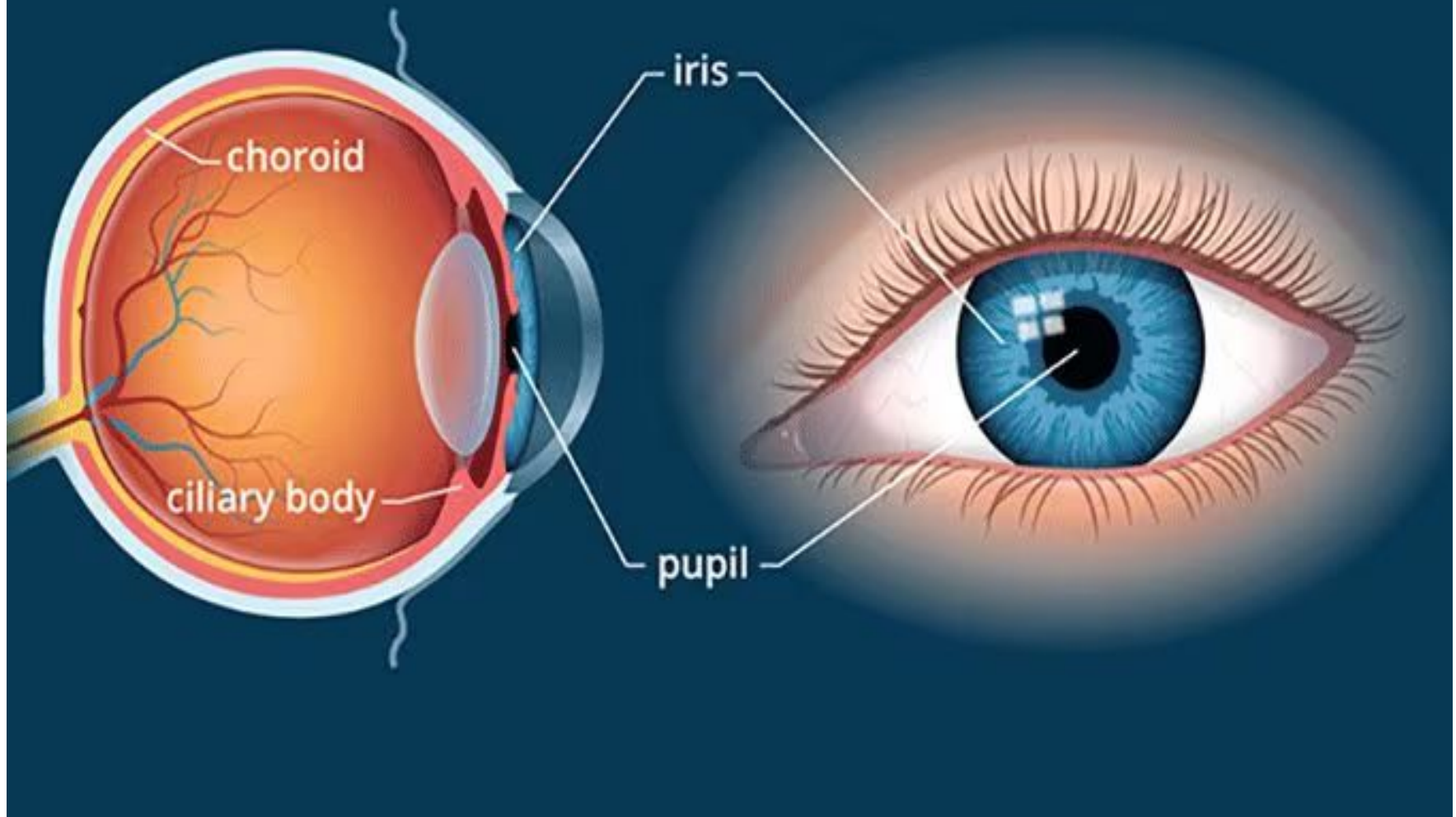


The middle coat:

- **Choroid:** is thin, pigmented layer which separates posterior part of sclera from the retina.
- Posteriorly it is perforated by optic nerve.
- It consists of the following layers:
 1. Suprachoroidal lamina
 2. Vascular lamina
 3. Choriocapillary lamina
 4. Membrane of Bruch

The middle coat:

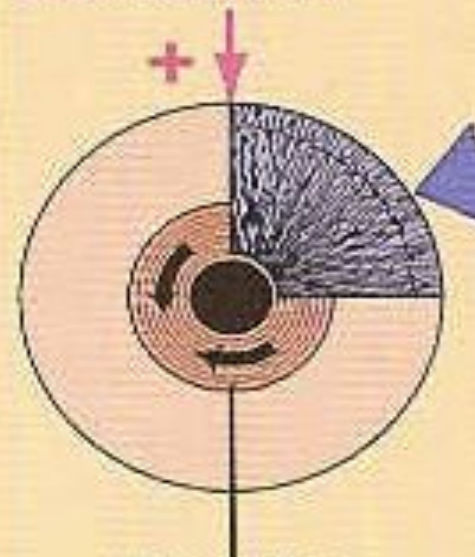
- **Ciliary body:** lies posterior to limbus and continuous anteriorly with the iris and posteriorly with the choroid.
- Its scleral surface contains ciliary muscle.
- The ciliary muscle is a ring of unstriated (smooth) muscle which are longitudinal, radial and circular.
- The ciliary muscle supports the lens and helps to modify shape of the lens.



The middle coat:

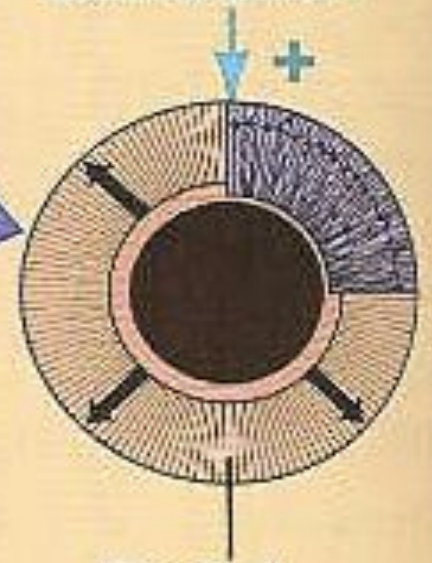
- **Iris:** is the anterior part of uveal tract and forms a circular curtain with an opening in the centre called **pupil**.
- By adjusting the size of pupil it controls the amount of light entering the eyeball, thus acting like a diaphragm.
- It is placed vertically between cornea and lens.
- Its peripheral margin is attached to the anterior surface of ciliary body and central free margin forms the boundary of pupil rests against the lens.
- Iris contains muscles called sphincter pupillae supplied by parasympathetic nerves and dilator pupillae supplied by sympathetic nerves.

Parasympathetic stimulation

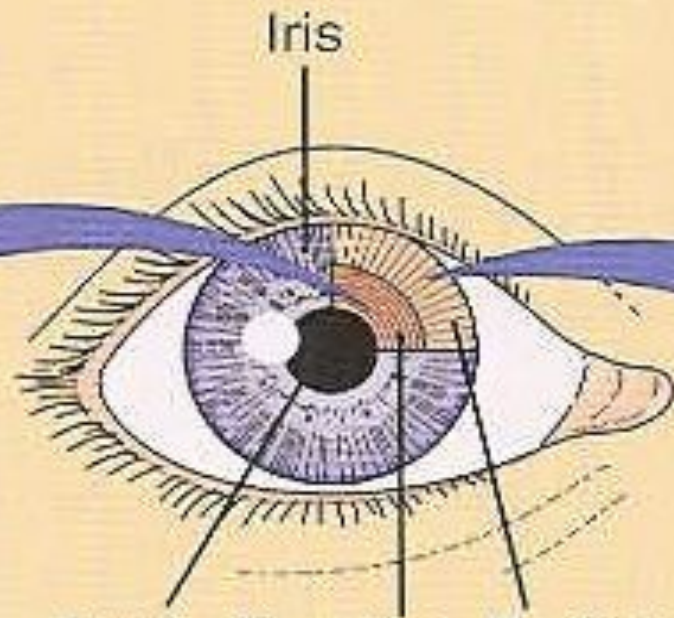


Circular (constrictor) muscle runs circularly

Sympathetic stimulation



Radial (dilator) muscle runs radially



Pupil
Circular muscle of iris
Radial muscle of iris

Pupillary constriction

Pupillary dilation

The inner coat:

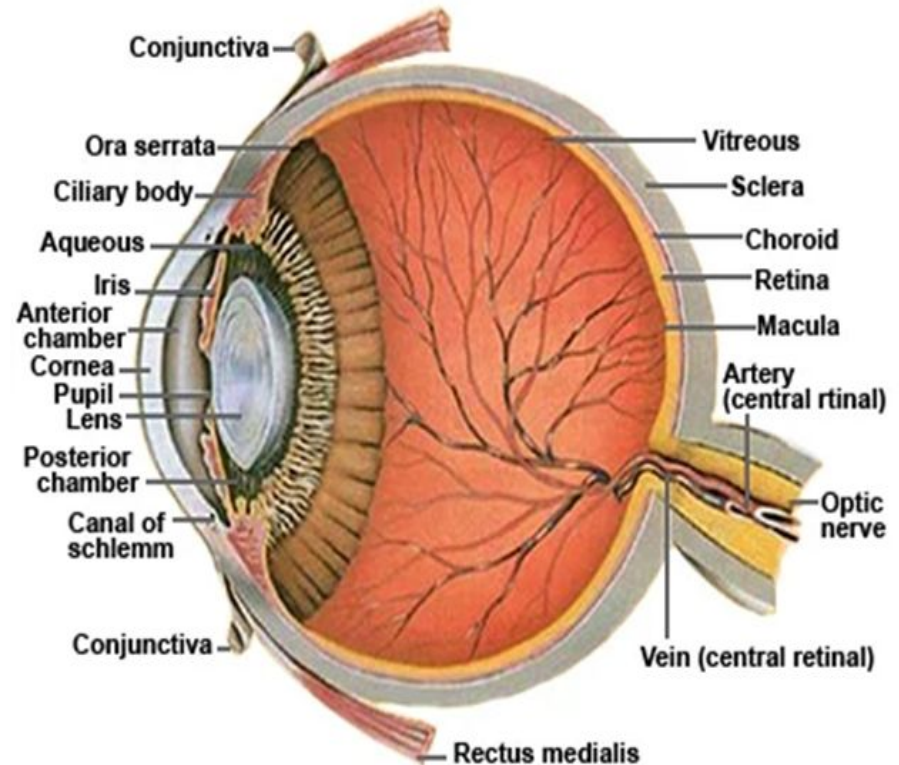
Retina: is the thin, delicate inner layer of eyeball which is continuous posteriorly with the optic nerve.

The retina is composed of ten layers:

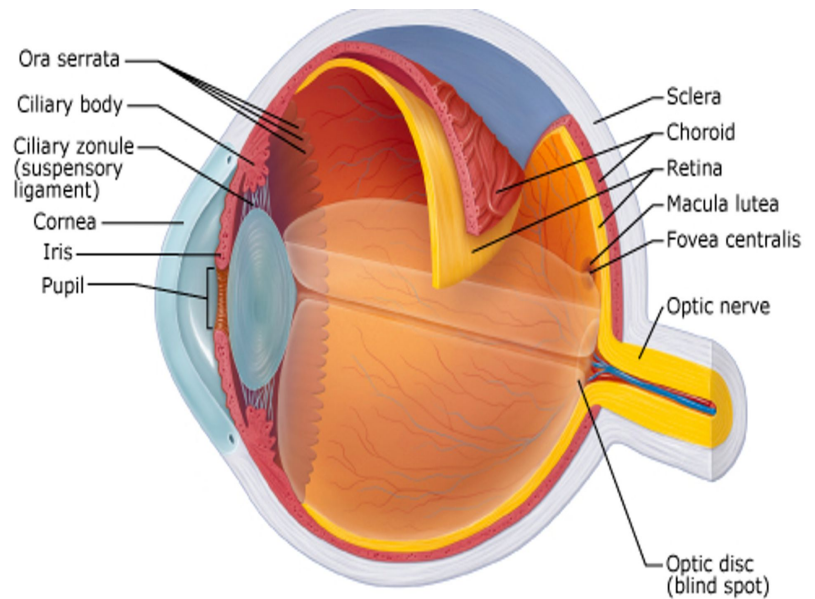
1. Outer pigmented layer
2. Layer of rods and cones
3. External limiting membrane
4. Outer nuclear layer
5. Outer plexiform layer
6. Inner nuclear layer
7. Inner plexiform layer
8. Ganglion cell layer
9. Nerve fibre layer
10. Internal limiting membrane

The inner coat:

- Opposite the area of entry of optic nerve there is a circular area called **optic disc**.
- The depressed area of optic nerve is called **physiological cup**. It contains no rods or cones therefore it is insensitive to light and termed as **physiological blind spot**.
- The rods and cones are light receptors of the eye. The rods respond to dim light and the cones respond to bright light.
- At the posterior pole of the eye, lateral to optic disc there is another depression called **macula lutea**. The center of macula is depressed to form **fovea centralis**. It contains cones only.
- The retina is supplied by central artery.

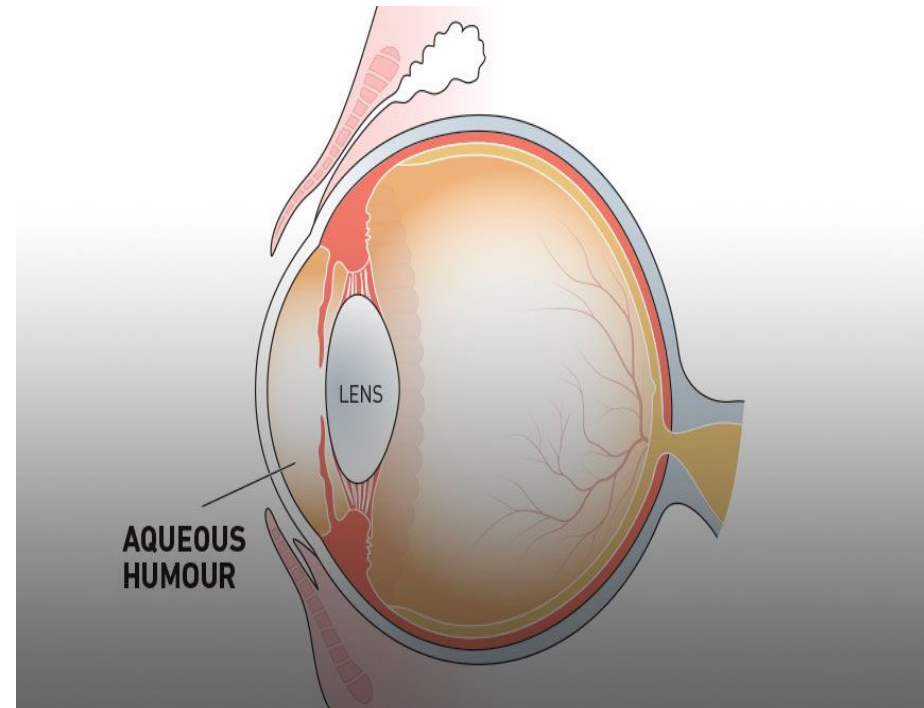


- **Ora Serrata:** is the serrated junction between the retina (optic part) and the ciliary body. This junction marks the transition from the simple non-photosensitive area of the retina to the complex, multi-layered photosensitive region.



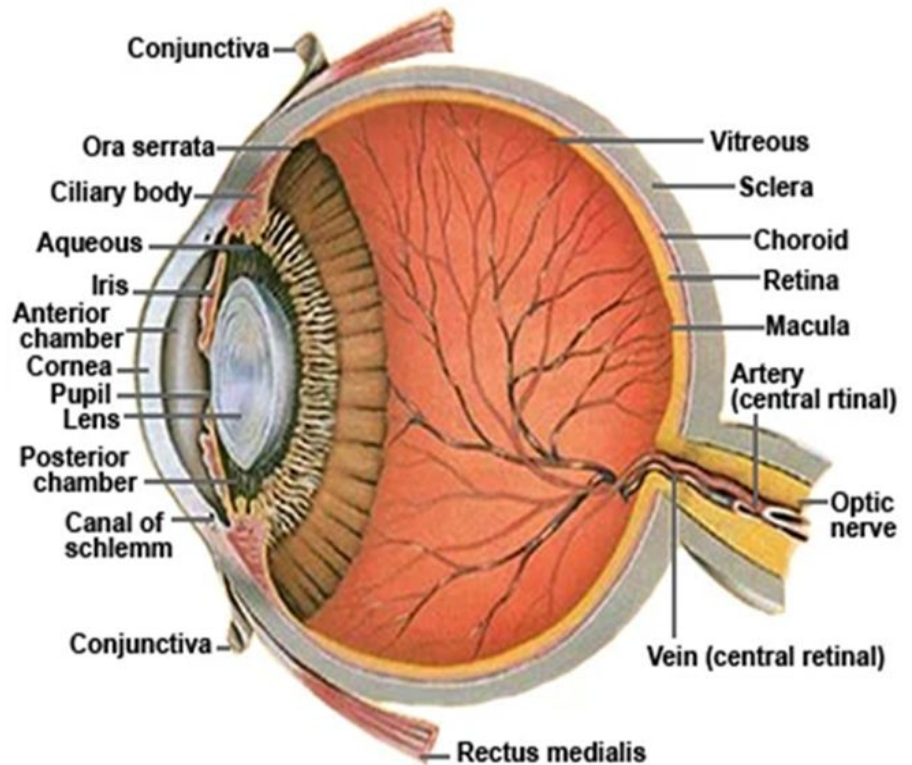
Aqueous humour:

- It is a clear fluid which fills the anterior segment.
- It is drained into anterior ciliary veins through canal of Schlemm.
- The intraocular pressure is mainly due to aqueous humor which maintains consistency & optical dimensions of the eyeball.
- Any impairment in the drainage of aqueous humor into the canal of Schlemm results in increased intraocular pressure called glaucoma.
- It contains ascorbic acid, glucose and amino acids and nourishes avascular tissues of cornea and lens.



Lens:

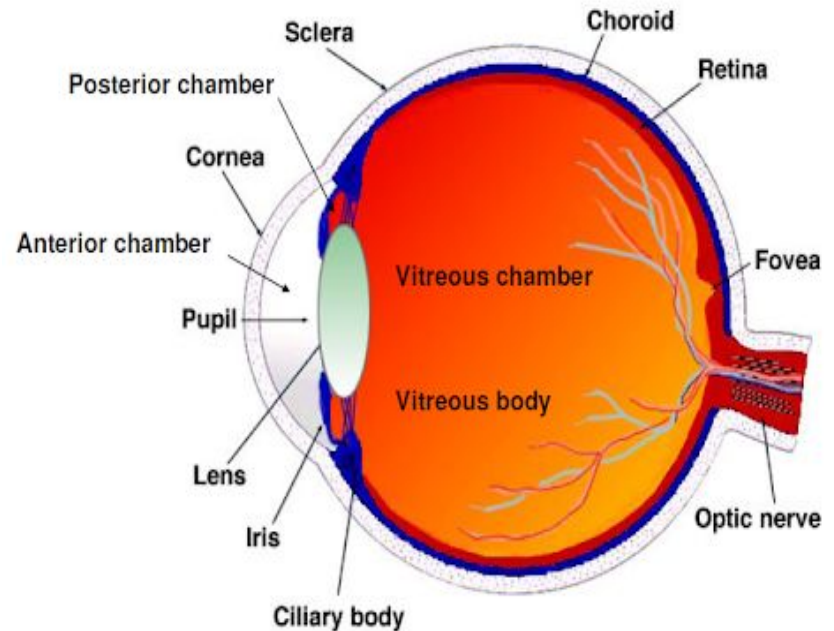
- **Lens:** is a transparent biconvex structure placed between anterior and posterior segments of the eye.
- It has power of 15 dioptries.
- The lens is enclosed in transparent elastic capsule which is thickest anteriorly near the circumference. Deep to capsule, anterior surface of lens is covered by capsular epithelium.
- The suspensory ligament of lens holds the lens in position and keeps its anterior surface flat.



Vitreous body:

- It is a colorless, jelly-like transparent mass which fills the posterior segment of the eye.
- It is enclosed in a delicate homogenous hyaloid membrane.

The vitreous chamber

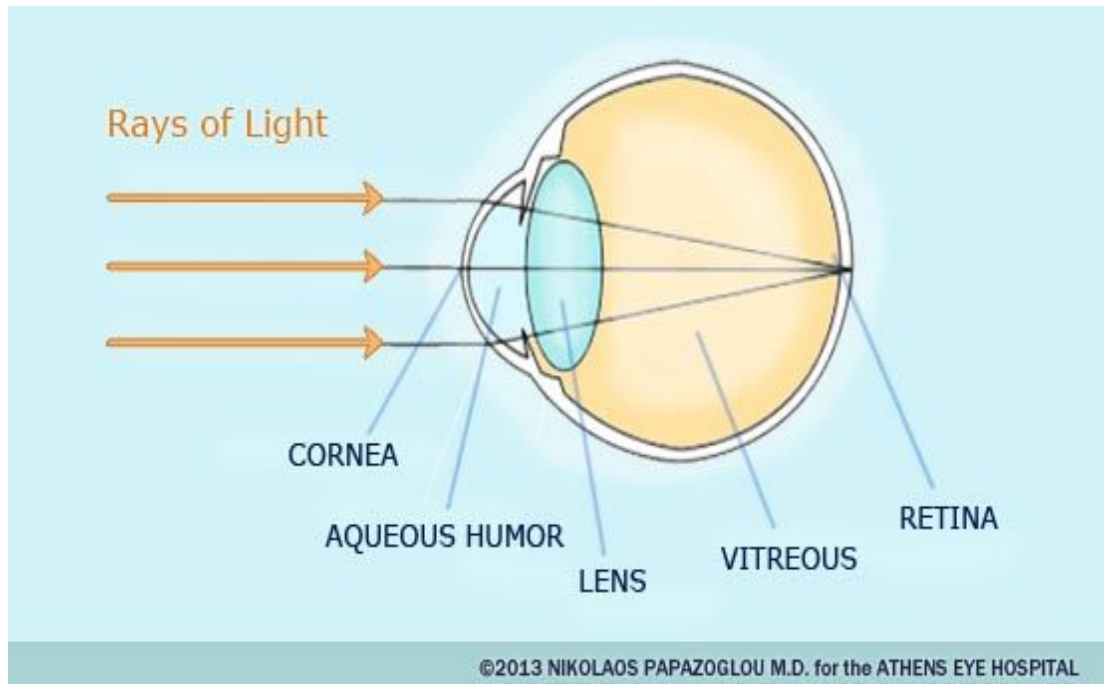


Refractive Media of the Eye:

- The refractive media of the eye are the structures that help in focusing the ray of light onto the retina where it can be detected by the photoreceptors.

- The human eye has **four** refractive media:

- 1) *Cornea,*
- 2) *Aqueous humor,*
- 3) *lens, and*
- 4) *Vitreous body*

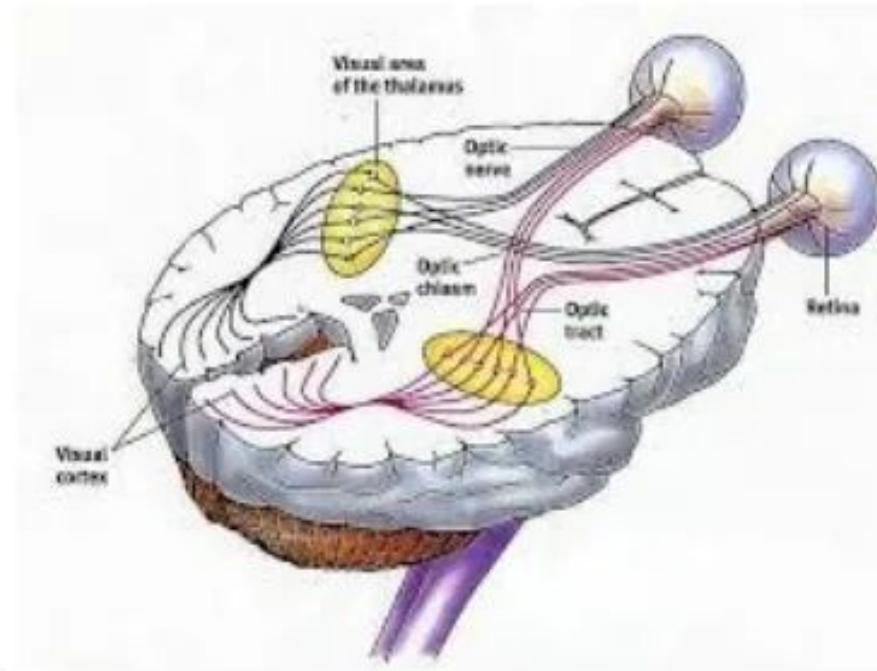


The Optic Nerve

The second cranial nerve, carries information from the retina to the brain.

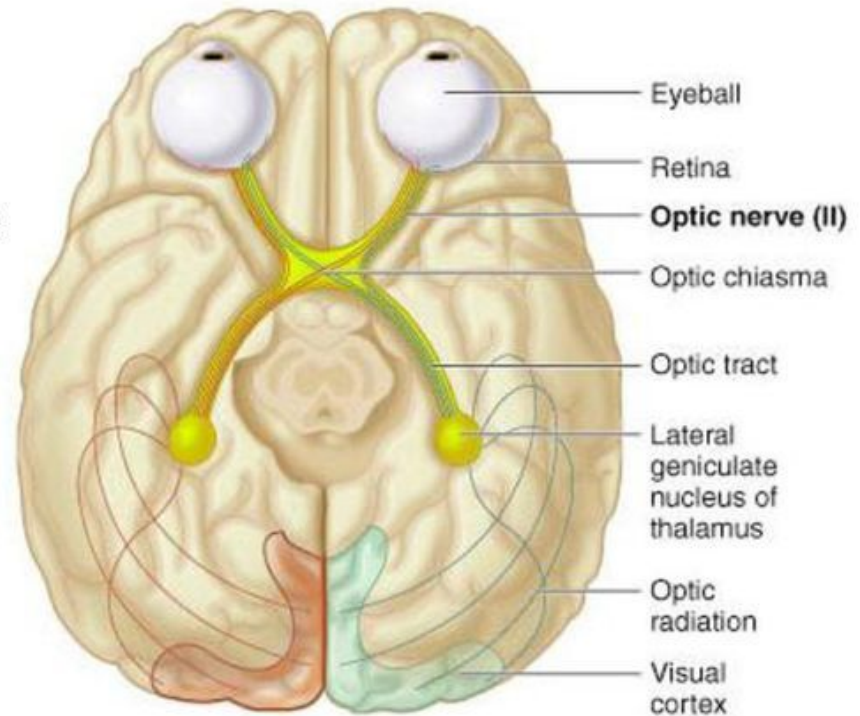
PATHWAY:

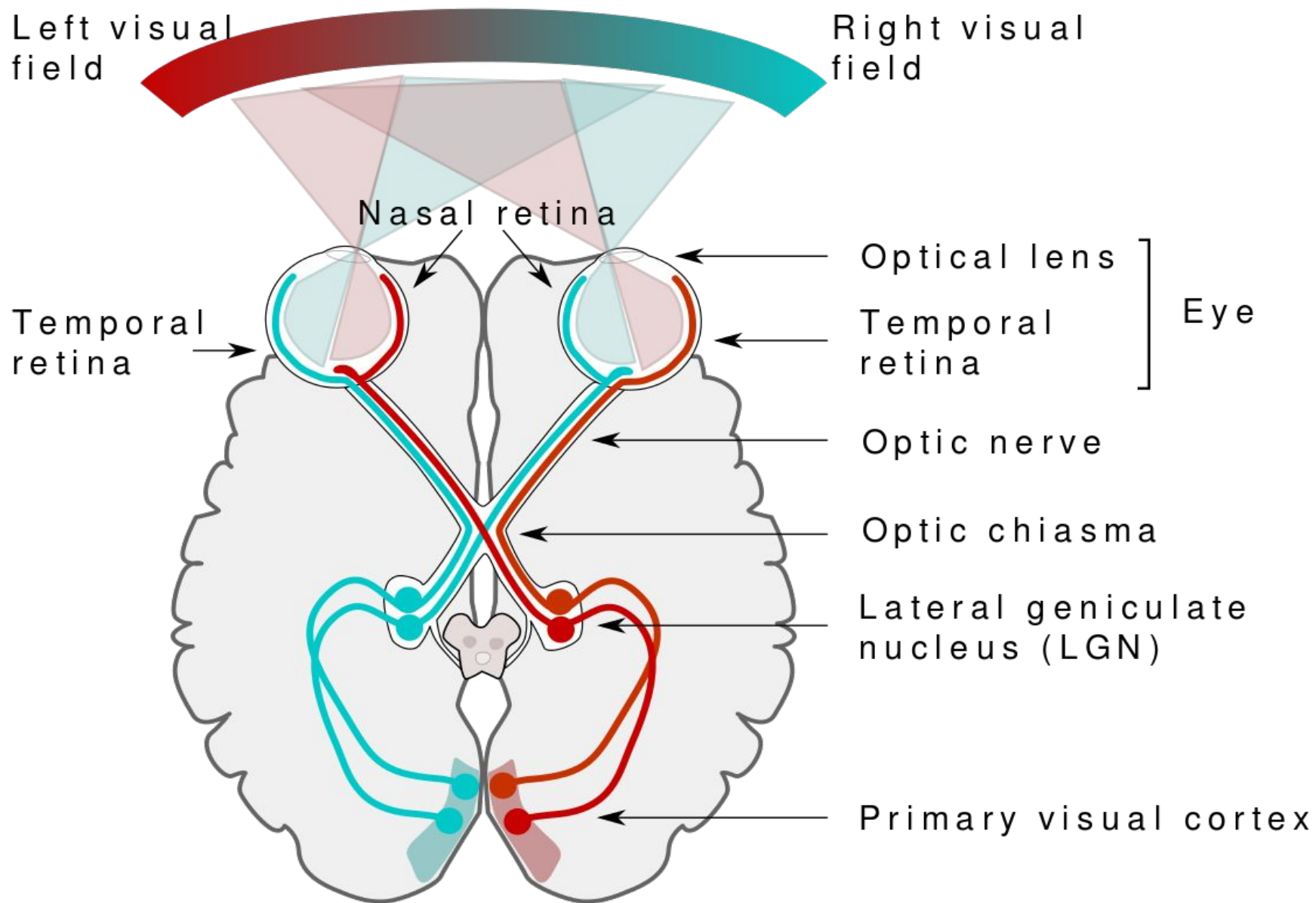
It is located in the back of the eye and leaves via the optic foramen, along with other optic vessels, as a continuation of the retinal ganglion cells. It travels from there into the optic chiasma, where fibers of opposite sides cross. From there, the nerve fibers form the optic tract which passes through the thalamus and then forms the optic radiation which continues until the visual cortex of the brain is reached. That is located in the occipital lobe. The visual cortex interprets signals brought in by the nerve

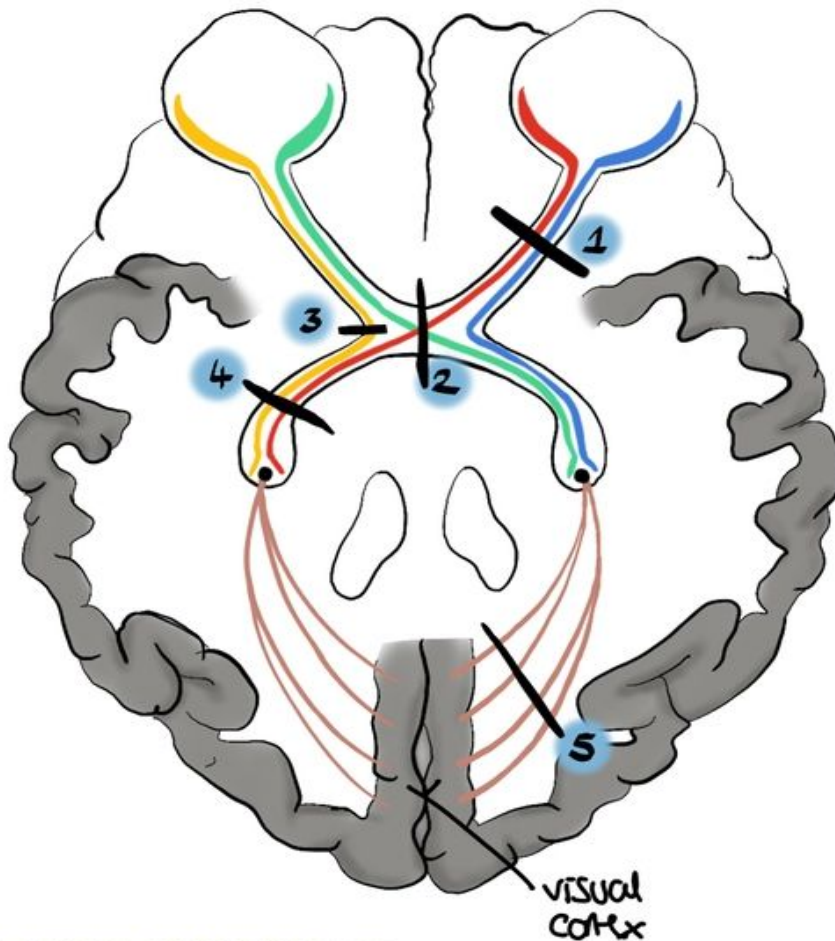


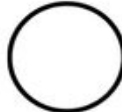
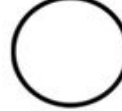
Cranial Nerve II: Optic

- Arises from the retina of the eye
- Optic nerves pass through the optic canals and converge at the optic chiasm
- They continue to the thalamus where they synapse
- From there, the optic radiation fibers run to the visual cortex
- Functions solely by carrying afferent impulses for vision







- | | | | |
|---|--|--|---|
| 1 |  |  | Total Right eye blindness |
| 2 |  |  | Bilateral hemianopia |
| 3 |  |  | Left nasal hemianopia |
| 4 |  |  | Right homonymous hemianopia |
| 5 |  |  | Left homonymous hemianopia with macular sparing |